

Tensix Neo™ IP

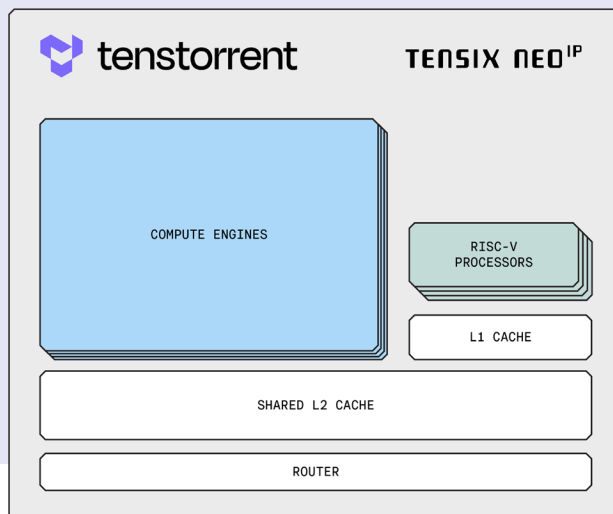
Optimized for Extraordinary AI Performance

Tensix NEO™ is Tenstorrent's latest high performance AI architecture, optimized for performance-per-mm² and performance-per-watt. The main unit of compute is the Tensix NEO Cluster, comprised of four Tensix NEO Cores, 4MB of shared L1 SRAM, high-bandwidth on-chip network, and data movement cores handling external DRAM and Ethernet I/O. Micro-architectural optimizations maximize FPU utilization and performance in the Tensix NEO Cores while updating the ISA to support more complex data patterns and modern data formats like the MX formats and FP4.

Toolchain and Support

- Open source simulation models available
- Python and C++ entry points for operators, with paths for kernel customization and optimization
- Powered by open source software:
 - The TT-Forge™ AI compiler allows for rapid model bring up and automatic optimization
 - The TT-Metalium™ SDK has a bottom-up approach that provides access directly to the metal, enabling use of Python and C++ for a variety of workloads
 - The TT-LLK SDK allows for tensor level programming

Metrics	Value
Technology	SF4X
Area	8.97 mm ² @ 1.4GHz
Frequency	
AI Clock	1.4GHz @ 0.75V
NoC Clock	1.38GHz @ 0.75V
Power (for BFLOAT16)	
	5.61W @ 1.4GHz, 0.75V, 85 OC, 60% FPU, 37% NoC Utilization
	7.98W @ 1.4GHz, 0.75V, 85 OC, 100% FPU, 50% NoC Utilization
Data formats (TFLOPS/Tensix)	
FP4/FP8	11.2 @ 1.4GHz
BFLOAT16	5.6 @ 1.4GHz
TF32/FP16	2.8 @ 1.4GHz
INT8	22.4 @ 1.4GHz
NoC	
Topology	2D Mesh
Bandwidth	256 B/cycles
SRAM Memory	4MB



Key Features

- Updated mesh architecture that is both power-efficient and user friendly
- Performant data movement through fine grained data shuffle
- Improved performance-per-watt and performance per mm²; each Tensix NEO cluster contains:
 - 4 Tensix NEO Cores
 - Shared 4MB L1 SRAM
 - Shared NoC and Data Movement Block
- Support for FP4 and MX data formats
- Parallel issue of vector and matrix instructions
- ASIL certified (NEO Auto only)

